3.1 p.167 #1, 5, 11, 20, 27, 33

1. $\begin{vmatrix} 3 & 0 & 4 \\ 2 & 3 & 2 \\ 0 & 5 & -1 \end{vmatrix} = 3\begin{vmatrix} 3 & 2 \\ 5 & -1 \end{vmatrix} - 0 + 4\begin{vmatrix} 2 & 3 \\ 0 & 5 \end{vmatrix}$

$= 3(-3-10) + 4(10-0)$
$= 3(-13) + 4(10)$
$= 1$

$= 0+3(-3-0)-5(6-8)$
$= 3(-3)-5(-2)$
$= 1$

5. $\begin{vmatrix} 2 & 3 & -4 \\ 4 & 0 & 5 \\ 5 & 1 & 6 \end{vmatrix} = 2(0-5) - 3(24-25) + (-4)(4-0)$
$= 2(-5) - 3(-1) - 4(4) = -23$

11. $\begin{vmatrix} 3 & 5 & -8 & 4 \\ 0 & -2 & 3 & -7 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 2 \end{vmatrix} = -0+0-0+2\begin{vmatrix} 3 & 5 & -8 \\ 0 & -2 & 3 \\ 0 & 0 & 1 \end{vmatrix} = 2\left(0-0+1\begin{vmatrix} 3 & 5 \\ 0 & -2 \end{vmatrix}\right)$

$= 2(1)(-6-0)$
$= -12$

20. $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \sim \begin{bmatrix} a & b \\ kc & kd \end{bmatrix}$ determinand multiplied by k b/c $kR_2 = R_2$

27. $A = \begin{bmatrix} k & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\det A = k(1)(1) = k$

33. $\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 0-1 = -1$ "$\det E$ $\quad \det A = ad-bc$

$EA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ a & b \end{bmatrix}$ $\quad \det EA = bc-ad$

$(\det E)(\det A) = -1(ad-bc) = bc-ad$

3.2 p. 175 #5, 11, 19, 23, 25

5. $\begin{vmatrix} 1 & 5 & -6 \\ -1 & -4 & 4 \\ -2 & -7 & 9 \end{vmatrix} \sim \begin{vmatrix} 1 & 5 & -6 \\ 0 & 1 & -2 \\ 0 & 3 & -3 \end{vmatrix} \sim 3\begin{vmatrix} 1 & 5 & -6 \\ 0 & 1 & -2 \\ 0 & 1 & -1 \end{vmatrix} \sim 3\begin{vmatrix} 1 & 5 & -6 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{vmatrix}$

$= 3$

11. $\begin{vmatrix} 2 & 5 & -3 & -1 \\ 3 & 0 & 1 & -3 \\ -6 & 0 & -4 & 9 \\ 4 & 10 & -4 & -1 \end{vmatrix} \sim \begin{vmatrix} 2 & 5 & -3 & -1 \\ 3 & 0 & 1 & -3 \\ 0 & 0 & -2 & 3 \\ 0 & 0 & 2 & 1 \end{vmatrix} \sim \begin{vmatrix} 2 & 5 & -3 & -1 \\ 3 & 0 & 1 & -3 \\ 0 & 0 & -2 & 3 \\ 0 & 0 & 0 & 4 \end{vmatrix}$

$= -0 + 0 - 0 + (4)\begin{vmatrix} 2 & 5 & -3 \\ 3 & 0 & 1 \\ 0 & 0 & -2 \end{vmatrix} = (4)(-2)(0-15) = -8(-15)$

$= 120$

19. $\begin{vmatrix} a & b & c \\ 2d+a & 2e+b & 2f+c \\ g & h & i \end{vmatrix} = 2\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = 14$

23. $\begin{bmatrix} 2 & 0 & 0 & 8 \\ 1 & -7 & -5 & 0 \\ 3 & 8 & 6 & 0 \\ 0 & 7 & 5 & 4 \end{bmatrix} = 2\begin{vmatrix} -7 & -5 & 0 \\ 8 & 6 & 0 \\ 7 & 5 & 4 \end{vmatrix} - 8\begin{vmatrix} 1 & -7 & -5 \\ 3 & 8 & 6 \\ 0 & 7 & 5 \end{vmatrix}$

$= 2(4)(-42+40) - 8\left[-7(6+15) + 5(8+21)\right]$

$= 8(-2) - 8\left(-7(21) + 5(29)\right)$

$= -16 + 56(21) - 40(29)$

$= -16 + 1176 - 1160$

$= 0$

Not invertible

25. $\begin{bmatrix} 7 \\ -4 \\ -6 \end{bmatrix}, \begin{bmatrix} -8 \\ 5 \\ 7 \end{bmatrix}, \begin{bmatrix} 7 \\ 0 \\ -5 \end{bmatrix}$

lin. indep. if columns of invertible matrix determinant $\neq 0$

$$\begin{vmatrix} 7 & -8 & 7 \\ -4 & 5 & 0 \\ -6 & 7 & -5 \end{vmatrix} = 7(-25) + 8(20) + 7(-28+30)$$

$$= -175 + 160 + 14$$

$$= -1 \neq 0$$

so yes

3.3 p. 184 # 1*, 5, 9, 11*, 21, 23, 27, 29

1. $5x_1 + 7x_2 = 3$
$2x_1 + 4x_2 = 1$

$\begin{vmatrix} 5 & 7 \\ 2 & 4 \end{vmatrix} = 20 - 14 = 6$

$x_1 = \dfrac{\begin{vmatrix} 3 & 7 \\ 1 & 4 \end{vmatrix}}{6} = \dfrac{12-7}{6} = \dfrac{5}{6}$ $\quad x_2 = \dfrac{\begin{vmatrix} 5 & 3 \\ 2 & 1 \end{vmatrix}}{6} = \dfrac{5-6}{6} = \dfrac{-1}{6}$

5. $\begin{vmatrix} 2 & 1 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 2 \end{vmatrix} = 0 - 1\begin{vmatrix} 2 & 0 \\ -3 & 1 \end{vmatrix} + 2\begin{vmatrix} 2 & 1 \\ -3 & 0 \end{vmatrix} = -1(2) + 2(0+3)$

$= -2 + 6$

$= 4$

$\begin{vmatrix} 7 & 1 & 0 \\ -8 & 0 & 1 \\ -3 & 1 & 2 \end{vmatrix} = -1\begin{vmatrix} -8 & 1 \\ -3 & 2 \end{vmatrix} + 0 - 1\begin{vmatrix} 7 & 0 \\ -8 & 1 \end{vmatrix} = -(-16+3) - 1(7-0)$

$= 13 - 7$

$= 6$

$x_1 = 6/4 = 3/2$

$\begin{vmatrix} 2 & 7 & 0 \\ -3 & -8 & 1 \\ 0 & -3 & 2 \end{vmatrix} = -1\begin{vmatrix} 2 & 7 \\ 0 & -3 \end{vmatrix} + 2\begin{vmatrix} 2 & 7 \\ -3 & -8 \end{vmatrix} = -(-6) + 2(-16+21)$

$= 6 + 2(5) = 16$

$x_2 = 16/4 = 4$

$\begin{vmatrix} 2 & 1 & 7 \\ -3 & 0 & -8 \\ 0 & 1 & -3 \end{vmatrix} = -\begin{vmatrix} -3 & -8 \\ 0 & -3 \end{vmatrix} - \begin{vmatrix} 2 & 7 \\ -3 & -8 \end{vmatrix} = -(9-0) - (-16+21)$

$= -9 - 5$

$= -14$

$x_3 = -14/4 = -7/2$

9. $\begin{vmatrix} s & -2s \\ 3 & 6s \end{vmatrix} = 6s^2 + 6s \neq 0$ $\qquad s \neq 0, -1$

$6s(s+1) \neq 0$

$\begin{vmatrix} -1 & -2s \\ 4 & 6s \end{vmatrix} = -6s + 8s = 2s$ $\qquad \begin{vmatrix} s & -1 \\ 3 & 4 \end{vmatrix} = 4s + 3$

$x_1 = \dfrac{2s}{6s^2+6s} = \dfrac{1}{3s+3}$ $\qquad x_2 = \dfrac{4s-3}{6s^2+6s}$

11. $\begin{bmatrix} 0 & -2 & -1 \\ 3 & 0 & 0 \\ -1 & 1 & 1 \end{bmatrix} = A$

$\det A = -3\begin{vmatrix} -2 & -1 \\ 1 & 1 \end{vmatrix}$

$= -3(-2+1) = 3$

$A^{-1} = \frac{1}{3}\begin{bmatrix} 0 & +1 & 0 \\ -3 & -1 & -3 \\ 3 & +2 & 6 \end{bmatrix}$

$= \begin{bmatrix} 0 & -1/3 & 0 \\ -1 & -1/3 & -1 \\ 1 & 2/3 & 2 \end{bmatrix}$

$C_{11} = \begin{vmatrix} 0 & 0 \\ 1 & 1 \end{vmatrix} = 0 \quad C_{21} = -\begin{vmatrix} -2 & -1 \\ 1 & 1 \end{vmatrix} = -(-2+1) = +1$

$C_{31} = \begin{vmatrix} -2 & -1 \\ 0 & 0 \end{vmatrix} = 0 \quad C_{12} = -\begin{vmatrix} 3 & 0 \\ -1 & 1 \end{vmatrix} = -3$

$C_{22} = \begin{vmatrix} 0 & -1 \\ -1 & 1 \end{vmatrix} = -1 \quad C_{32} = -\begin{vmatrix} 0 & -1 \\ 3 & 0 \end{vmatrix} = -3$

$C_{13} = \begin{vmatrix} 3 & 0 \\ -1 & 1 \end{vmatrix} = 3 \quad C_{23} = -\begin{vmatrix} 0 & -2 \\ -1 & 1 \end{vmatrix} = +2$

$C_{33} = \begin{vmatrix} 0 & -2 \\ 3 & 0 \end{vmatrix} = 6$

21. $(-1,0)\ (0,5)\ (1,-4)\ (2,1)$
$(0,0)\ (1,5)\ (2,-4)\ (3,1)$

$\begin{vmatrix} 1 & 2 \\ 5 & -4 \end{vmatrix} = -4-10 = -14$

area = 14

23. $\begin{vmatrix} 1 & 1 & 7 \\ 0 & 2 & 1 \\ -2 & 4 & 0 \end{vmatrix} = 2\begin{vmatrix} 1 & 7 \\ -2 & 0 \end{vmatrix} - \begin{vmatrix} 1 & 1 \\ -2 & 4 \end{vmatrix} = 2(0+14) - (4+2)$

$= 2(14) - 6$

$= 28 - 6$

$= 22$

volume = 22

27. $\det A = \begin{vmatrix} 6 & -2 \\ -3 & 2 \end{vmatrix} = 12 - 6 = 6$

area image = 24

$\text{area} = \begin{vmatrix} -2 & -2 \\ 3 & 5 \end{vmatrix} = |-10+6| = |-4| = 4$

29. $0, v_1, v_2$ in $\mathbb{R}^2$ → triangle → area

area $\triangle$ = $\frac{1}{2}$ area $\square$ = $\frac{1}{2}$ det $[v_1\ v_2]$